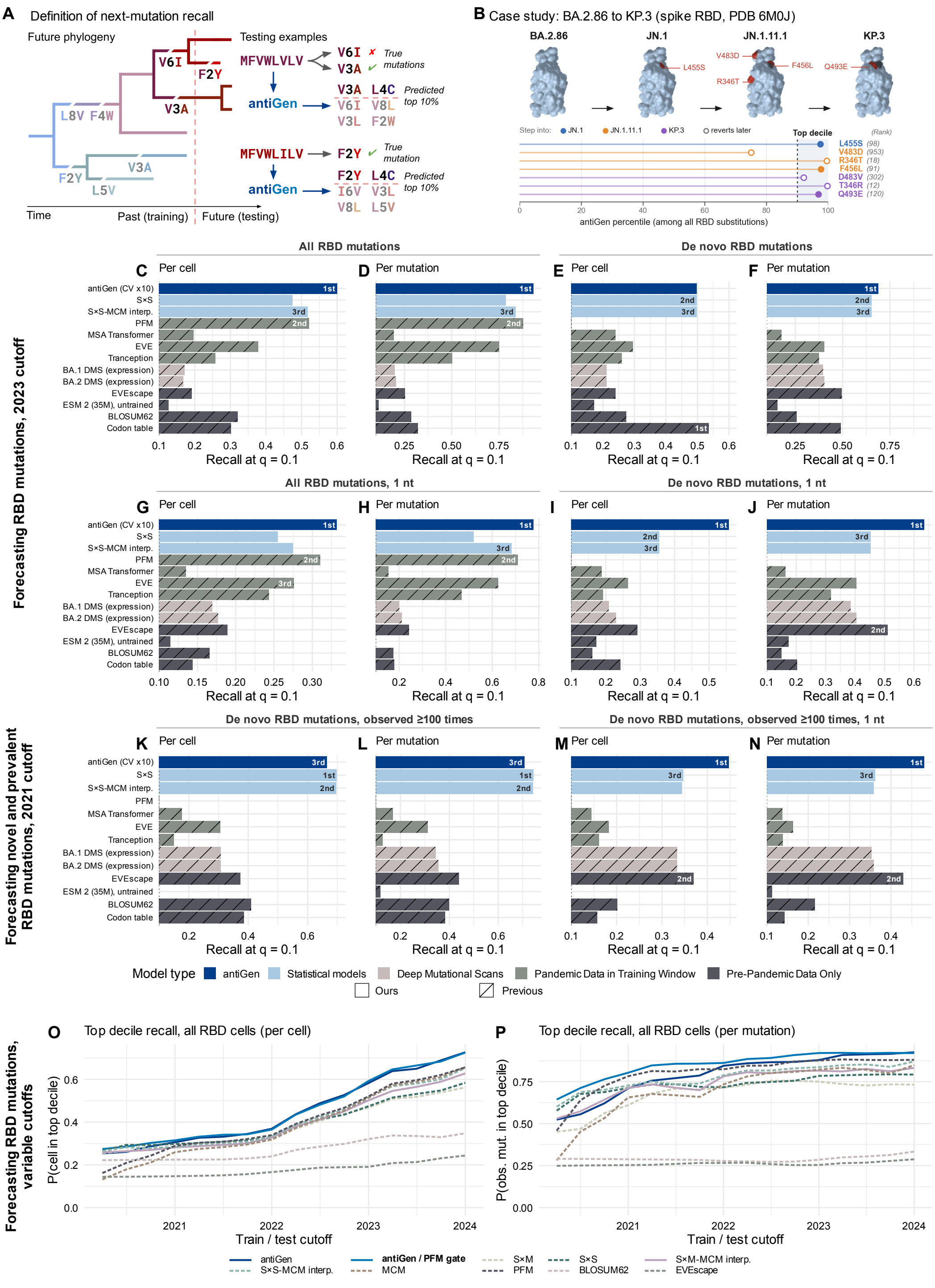

A
Definition of next-mutation recall
Future phylogeny
Testing examples
V6I
F2Y
V3A
L8V F4W
F2Y
V3A
L5V
MFVWLVLV
V6I ✗ True mutations
V3A ✓
antiGen
V3A L4C Predicted top 10%
V6I V8L
V3L F2W
MFVWLILV
F2Y ✓ True mutation
antiGen
F2Y L4C Predicted top 10%
I6V V3L
V8L L5V
Time
Past (training)
Future (testing)
B Case study: BA.2.86 to KP.3 (spike RBD, PDB 6M0J)
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
F456L
R346T
Q493E
Step into: JN.1 JN.1.11.1 KP.3 reverts later
Top decile
(Rank)
L455S (98)
V483D (953)
R346T (18)
F456L (91)
D483V (302)
T346R (12)
Q493E (120)
0 20 40 60 80 100
antiGen percentile (among all RBD substitutions)
All RBD mutations
De novo RBD mutations
C Per cell
D Per mutation
E Per cell
F Per mutation
Forecasting RBD mutations, 2023 cutoff
antiGen (CV x10)
S×S
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
Recall at q = 0.1
All RBD mutations, 1 nt
De novo RBD mutations, 1 nt
G Per cell
H Per mutation
I Per cell
J Per mutation
Forecasting novel and prevalent RBD mutations, 2021 cutoff
De novo RBD mutations, observed ≥100 times
De novo RBD mutations, observed ≥100 times, 1 nt
K Per cell
L Per mutation
M Per cell
N Per mutation
Model type
antiGen
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
Forecasting RBD mutations, variable cutoffs
O Top decile recall, all RBD cells (per cell)
P Top decile recall, all RBD cells (per mutation)
P(cell in top decile)
P(obs. mut. in top decile)
Train / test cutoff
2021
2022
2023
2024
antiGen
antiGen / PFM gate
S×M
S×M-MCM interp.
S×S-MCM interp.
MCM
S×S
PFM
BLOSUM62
EVEscape